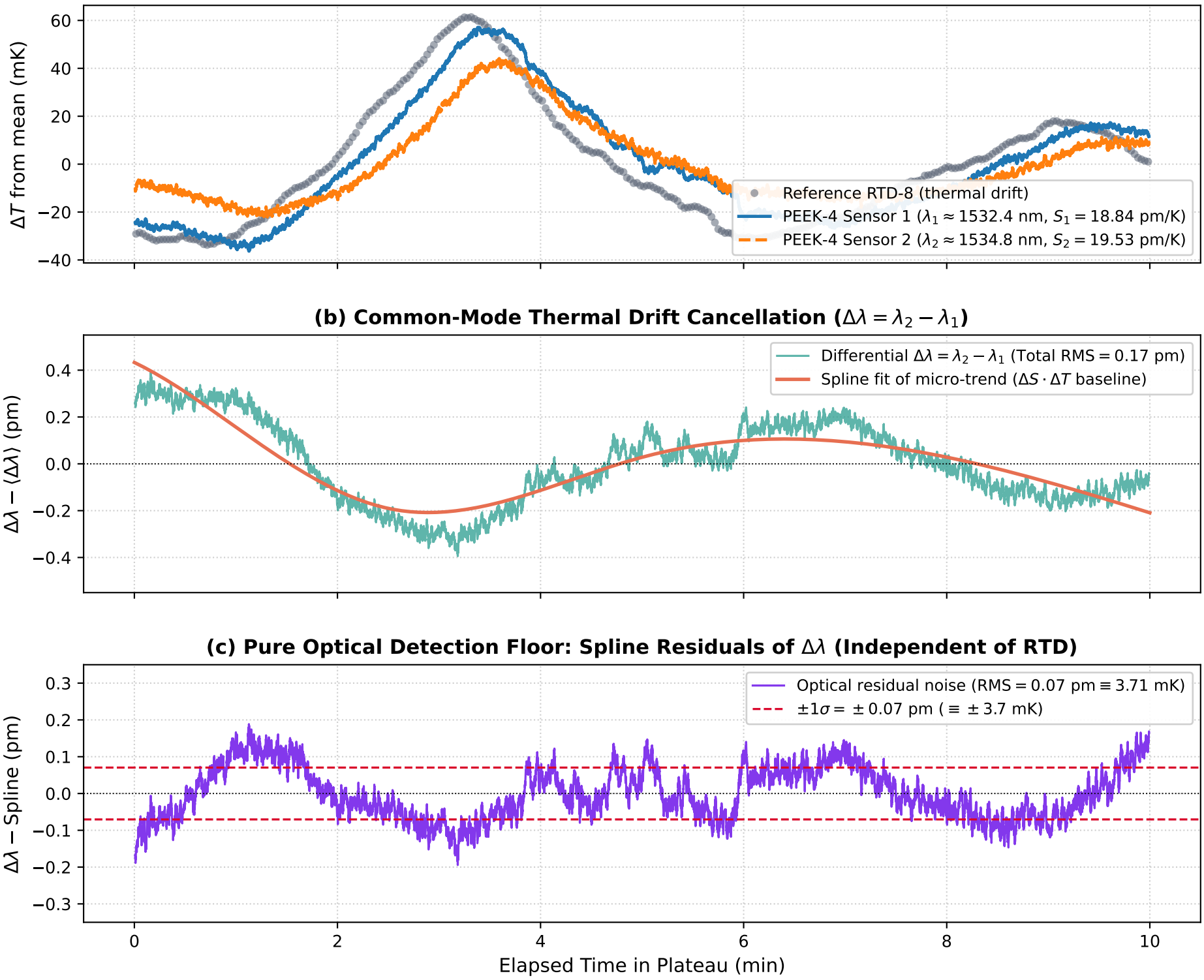

(a) Simultaneous Thermal Tracking in PEEK-4 Array (Plateau P3_R3: Total Drift ≈ 90 mK)
ΔT from mean (mK)
60
40
20
0
−20
−40
Reference RTD-8 (thermal drift)
PEEK-4 Sensor 1 (λ1 ≈ 1532.4 nm, S1 = 18.84 pm/K)
PEEK-4 Sensor 2 (λ2 ≈ 1534.8 nm, S2 = 19.53 pm/K)
(b) Common-Mode Thermal Drift Cancellation (Δλ = λ2 − λ1)
Δλ − ⟨Δλ⟩ (pm)
0.4
0.2
0.0
−0.2
−0.4
Differential Δλ = λ2 − λ1 (Total RMS = 0.17 pm)
Spline fit of micro-trend (ΔS · ΔT baseline)
(c) Pure Optical Detection Floor: Spline Residuals of Δλ (Independent of RTD)
Δλ − Spline (pm)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Optical residual noise (RMS = 0.07 pm ≡ 3.71 mK)
±1σ = ± 0.07 pm (≡ ± 3.7 mK)
0
2
4
6
8
10
Elapsed Time in Plateau (min)